EECS 484, Winter 2025- Database Management Systems

Tree-Structured Indexes

Chapter 10

Instructor: Barzan Mozafari

University of Michigan, Ann Arbor

Index Design Space

Organization Structure for k*

- Hash-based

  (+) Equality search
- Tree-based

  (+) Range, equality search
  - B+Tree (dynamic)
  - ISAM (static)

Data Entry (k*) Contents

1. Actual Data record

   index = file
2. <k, rid>

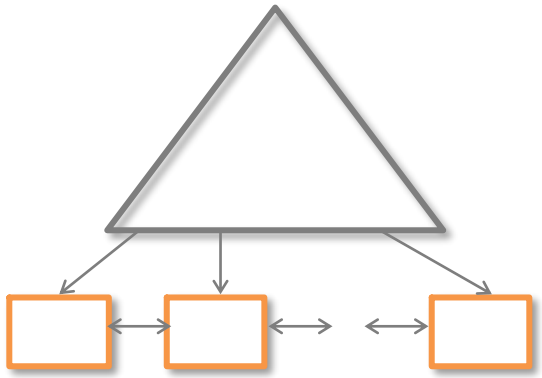
   actual records in a diff file
3. <k, list of rids>

Motivation

- Range and equality searches very common
- Can scan heap file, but this is expensive
- Goal: Create a dynamic index structure that allows for efficient evaluation, and equality / range queries.

B+ Tree

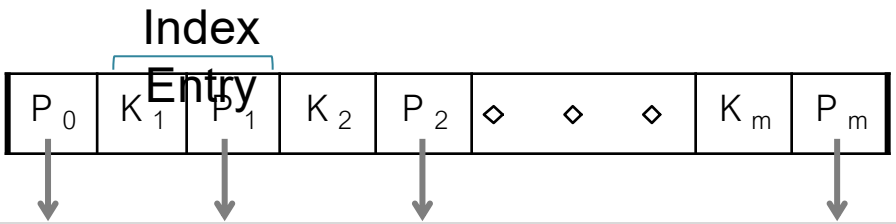
- Height-balanced (dynamic) tree search structure
- Supports equality and range-searches efficiently
- Main Characteristics:
  1. Minimum 50% occupancy (except for root), i.e. each node contains at least d entries, where d = order of the tree, i.e., d=ceiling(max number of entries / 2).
  2. Height of tree: the length of any path from the root to a leaf.
  3. Remains balanced after insert/delete operation.



Index entries
(direct search)

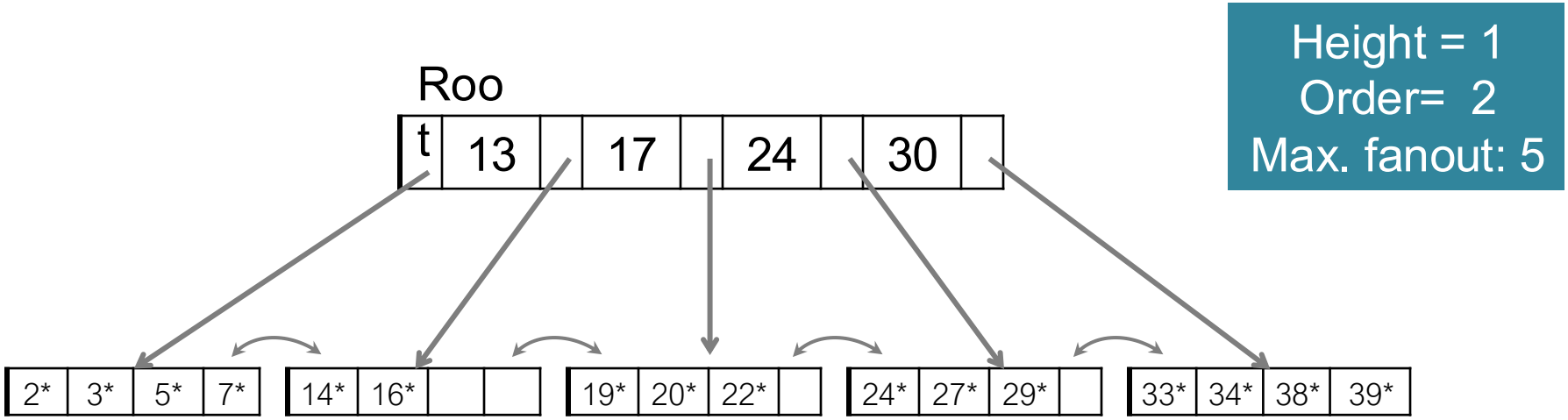
Data entries

Index Entries:
Entries in the non-leaf pages:
(search key value, pageid)

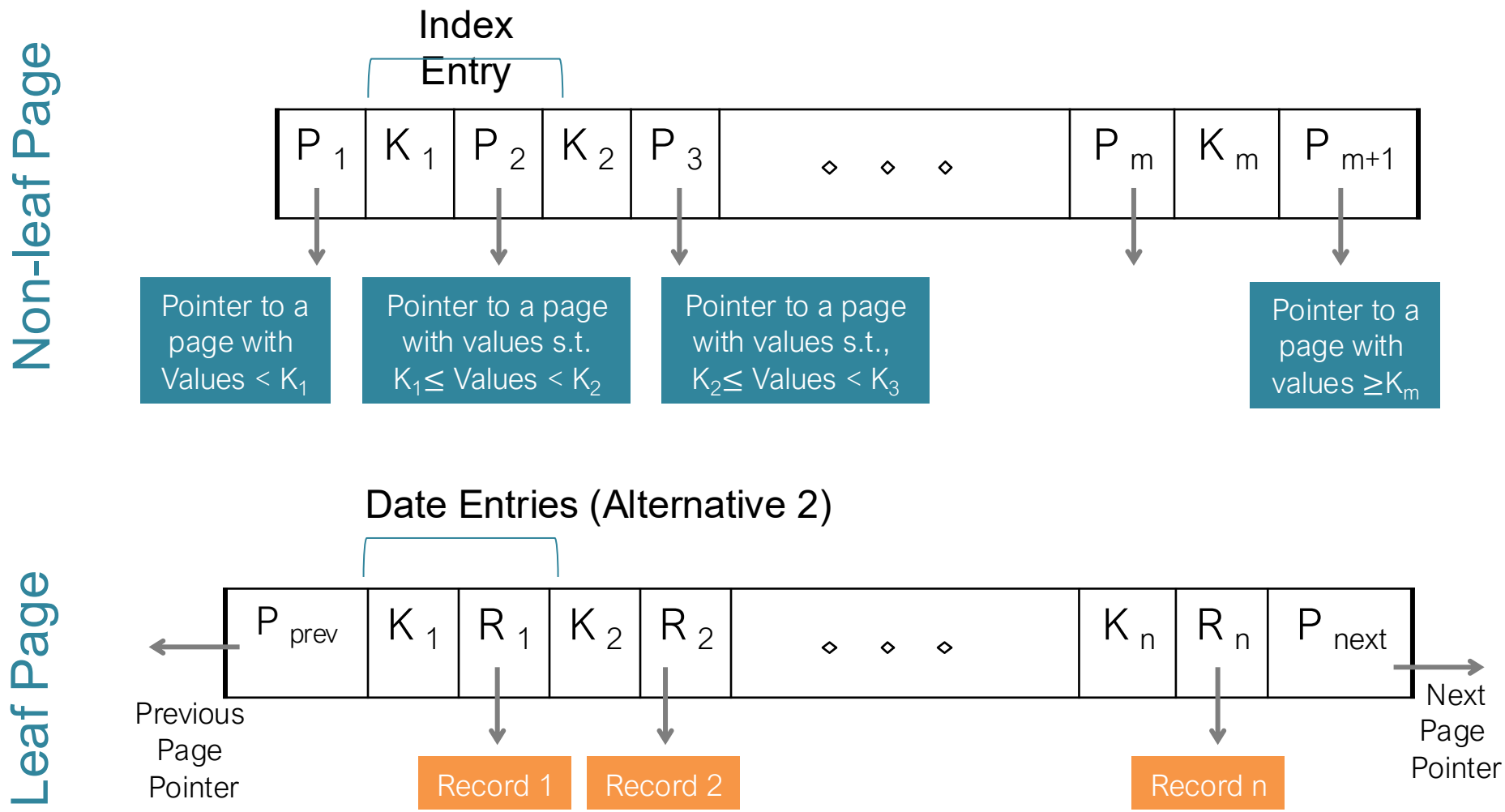


Searching in a B+ Tree

- Starting from root, examine index entries in non-leaf nodes, and traverse down the tree until a leaf node is reached
  - Non-leaf nodes can be searched using a binary or a linear search.
- Try it out: Search for 5*, 15*, all data entries >=24*



B+-Tree Page Format



B+ Trees: Height

What is the height of a B+ tree? (formula)

Definitions:

F: fanout (average number of children for non-leaf node)

N: total leaf pages

$\log_F N$

Quiz

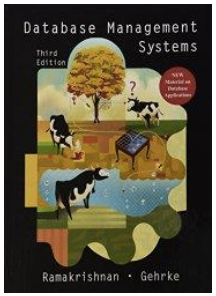
- You are given a file of 10 million records
- Suppose you can store 10 data entries per leaf page
- You build a B+ Tree with order 75, 67% average fill-factor

- What is the avg fanout?
- What is the height of your B+ Tree?

A Note on Order

- Some literature uses: order = max # of entries
- **In this class**: order = minimum # of entries

  (book uses this)



- *Order d* concept replaced by physical space criterion in practice (e.g. at least half-full)
  - Index (i.e. non-leaf) pages can typically hold many more entries than leaf pages because they only keep pointers and entries not the data entries

B+ Tree Operations

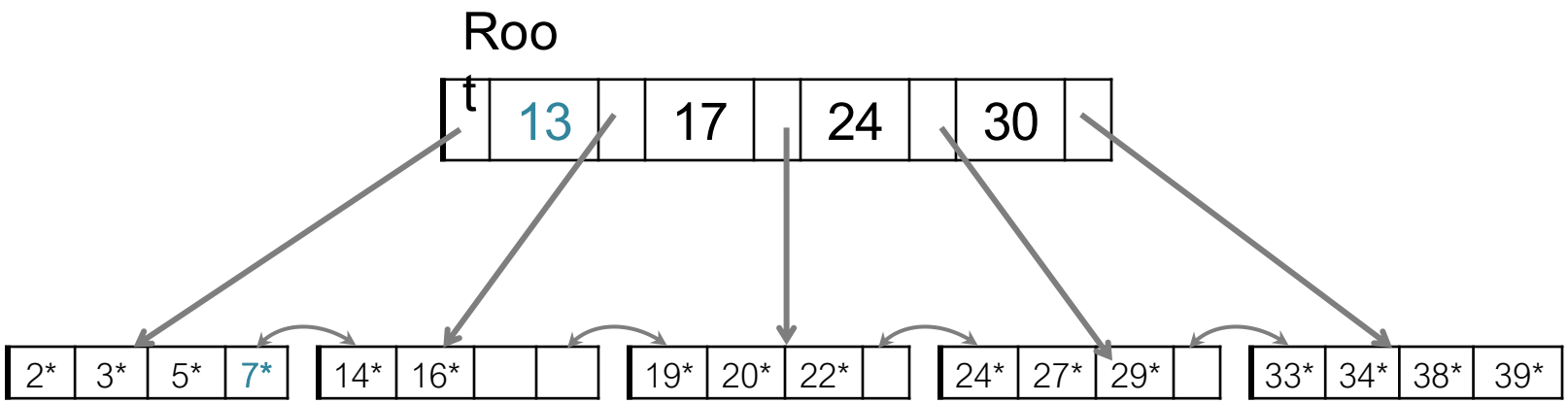
- Search
  - Equality
  - Range
- Insert data entry
- Delete data entry
- Bulk load

B+ Tree: Inserting a Data Entry

- Maintain invariants:
  - Search-tree property
  - All nodes must be at least ½ full (except root node), i.e., has between d and 2d entries
  - Root node is allowed to have a single entry

- Strategy:
  - Split nodes when they become full and a node is added:
    - Split an overfull node with (2d + 1) entries into two nodes with d and (d+1) entries, resp.

Inserting 8* into B+ Tree



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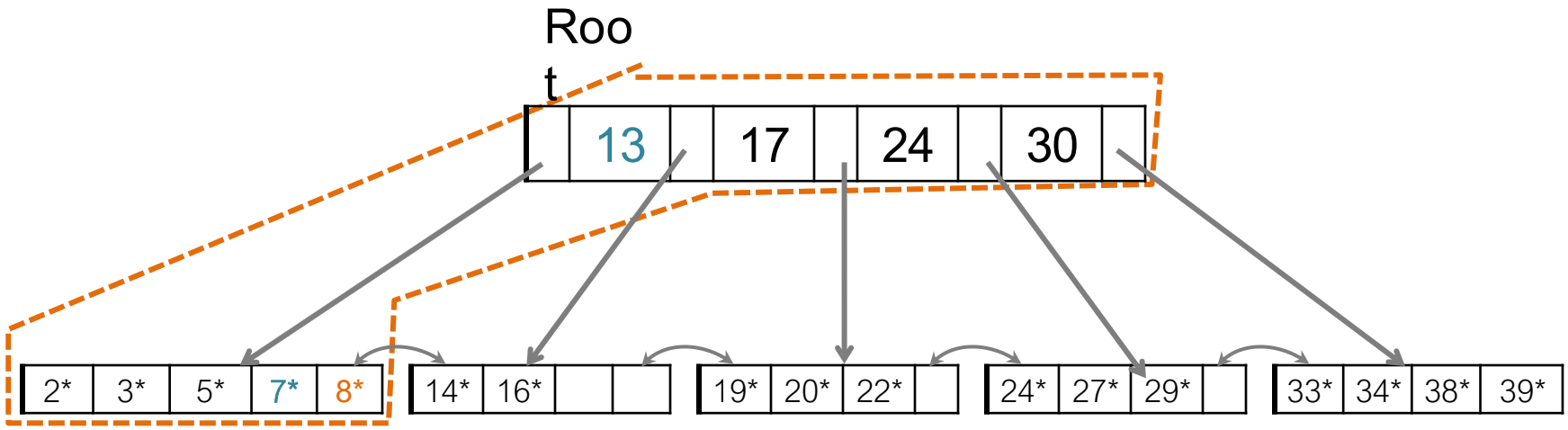
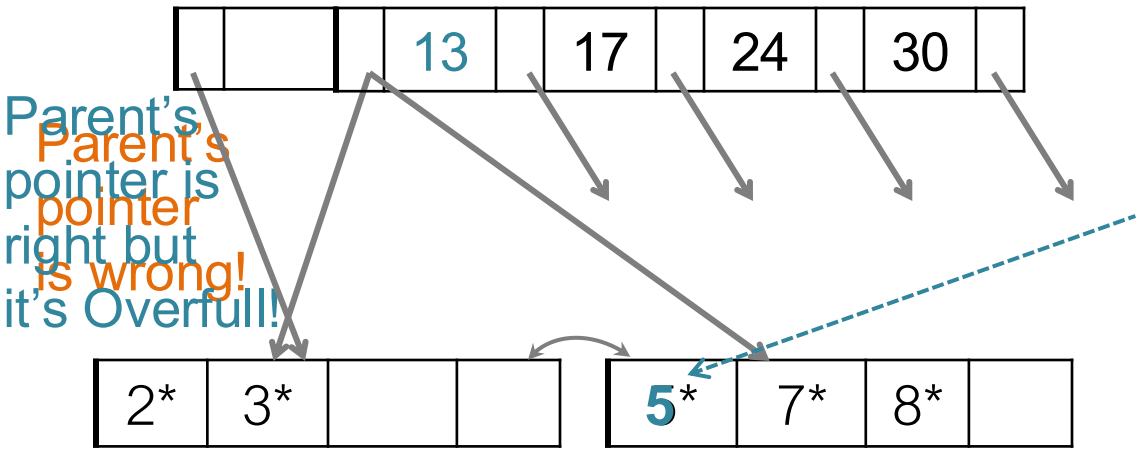


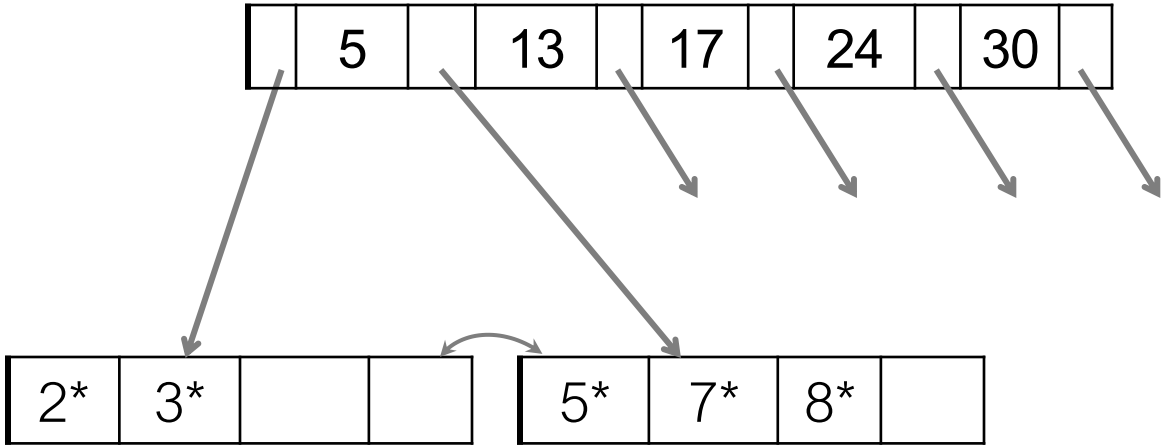
Overfull



Leaf split: The middle key is copied up to the parent (and continues to appear in the leaf)

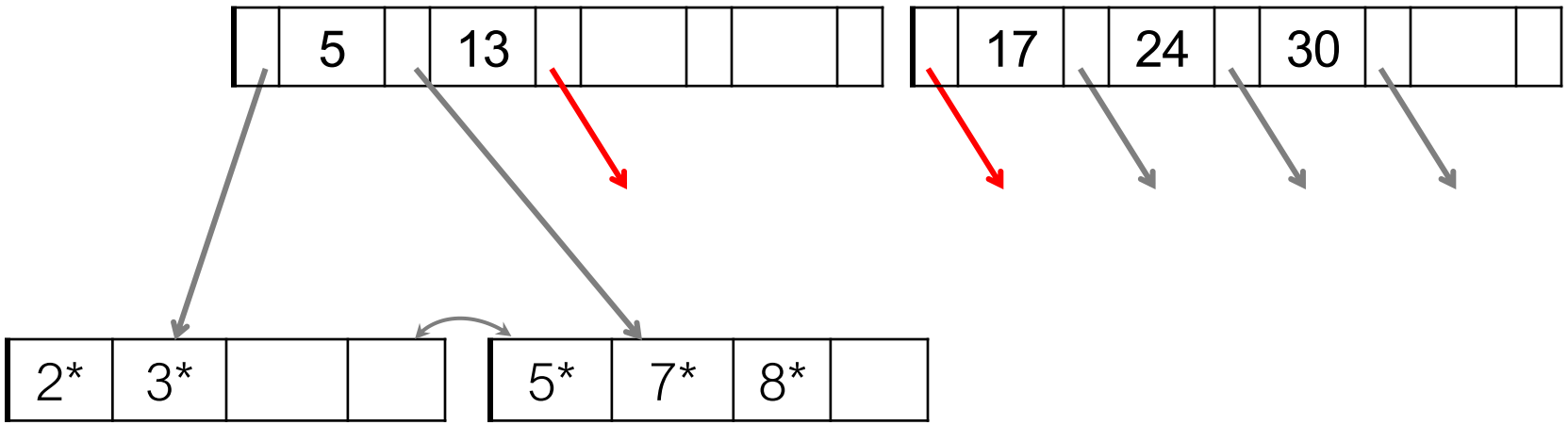
Splitting Overfull Non-Leaf Nodes

Overfull

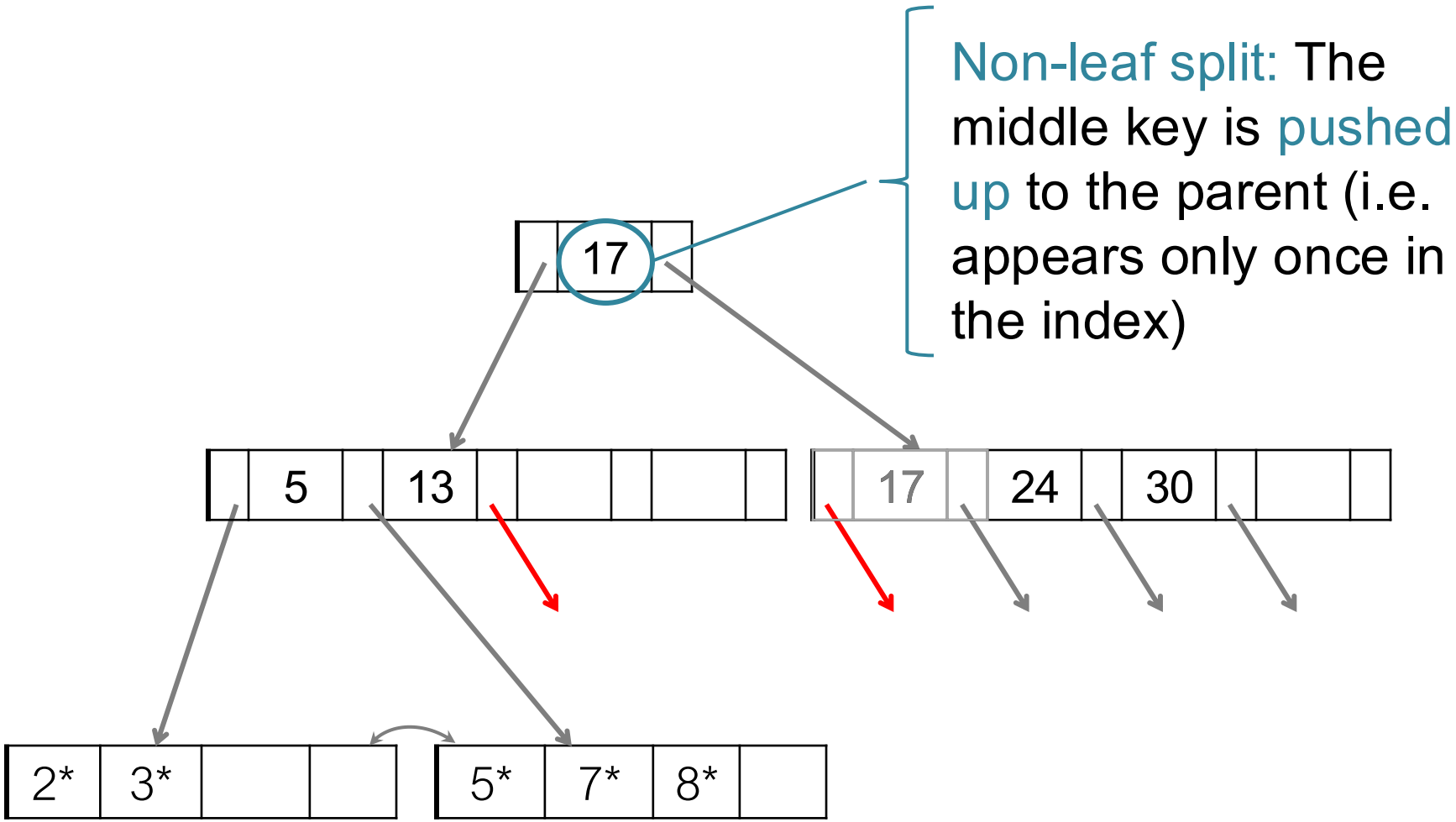


Splitting Overfull Non-Leaf Nodes

Split

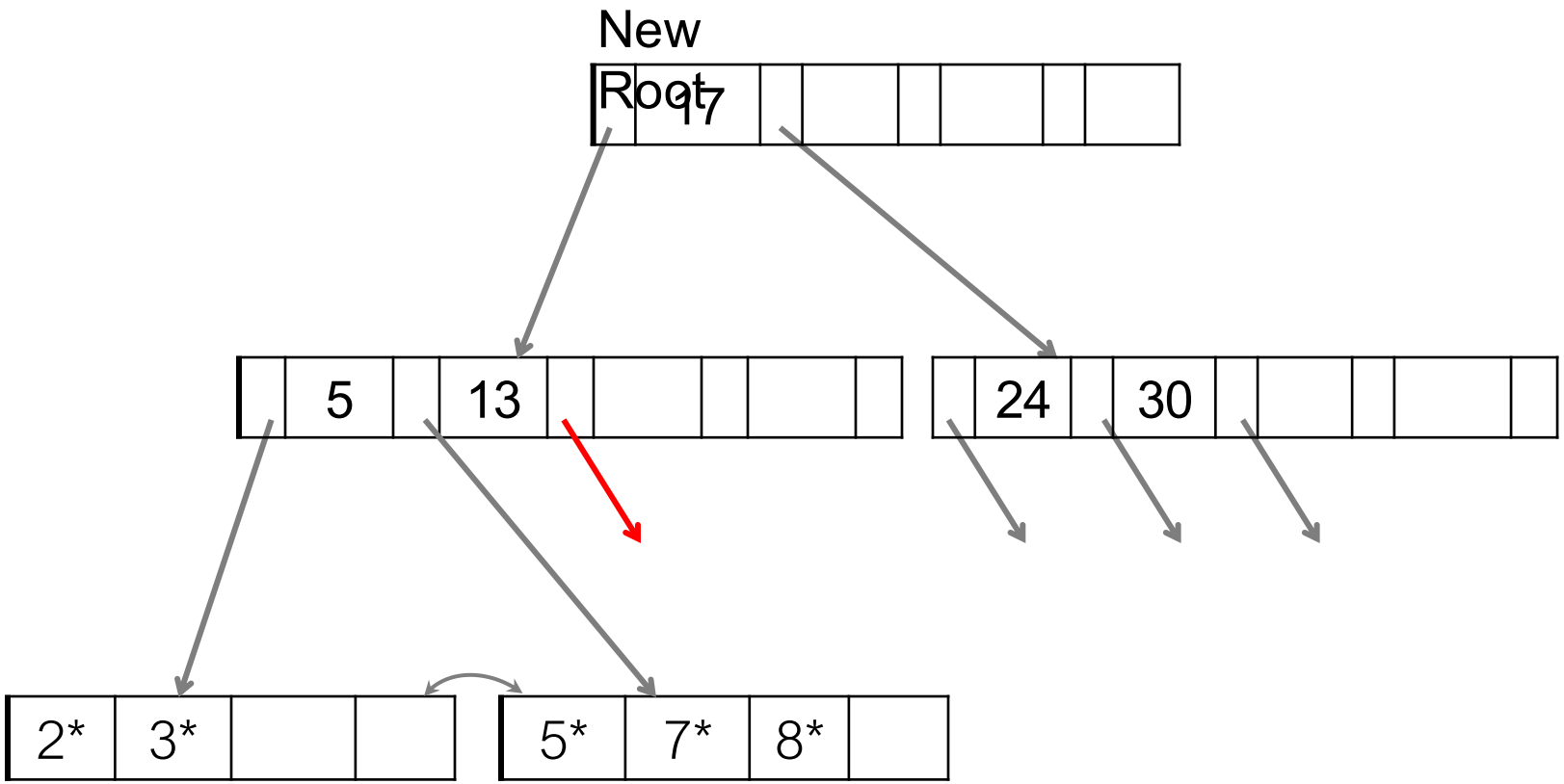


Splitting Overfull Non-Leaf Nodes

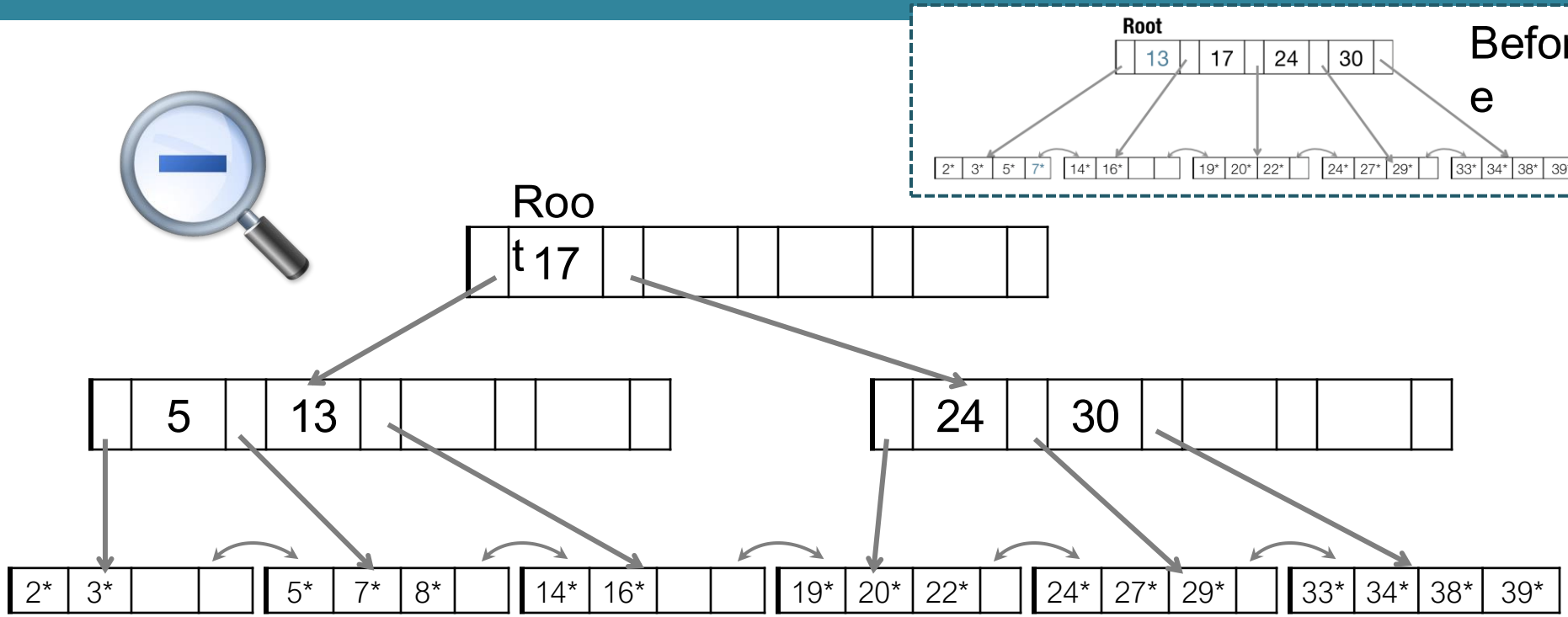


Splitting Overfull Non-Leaf Nodes

Minimum occupancy is guaranteed in both leaf and index page splits (but not in the root)

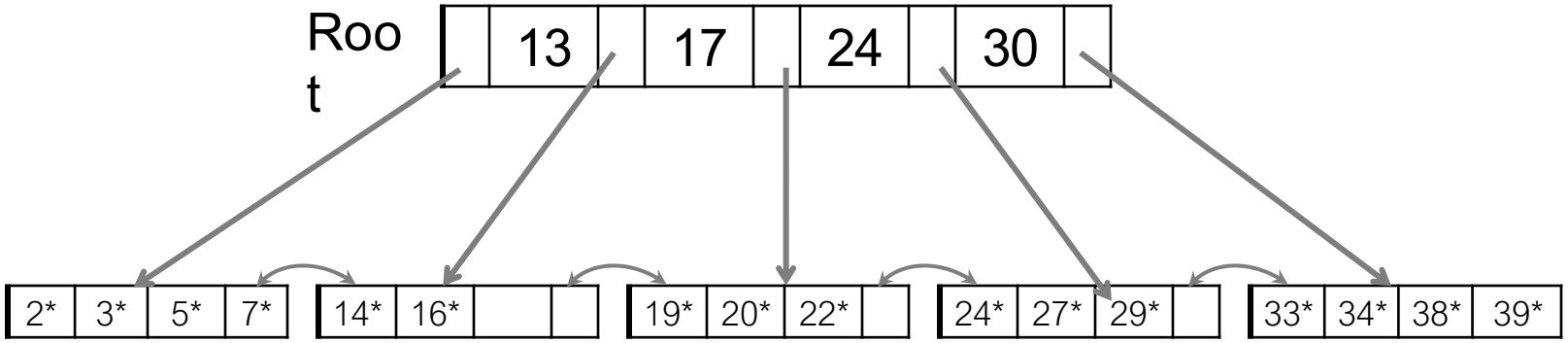


The B+-tree After Inserting 8*



- Root was split: height increases by 1
- Could avoid split by re-distributing entries with a sibling
  - Sibling: immediately to left or right AND same parent

Inserting 8* via Entry Re-distribution with Siblings



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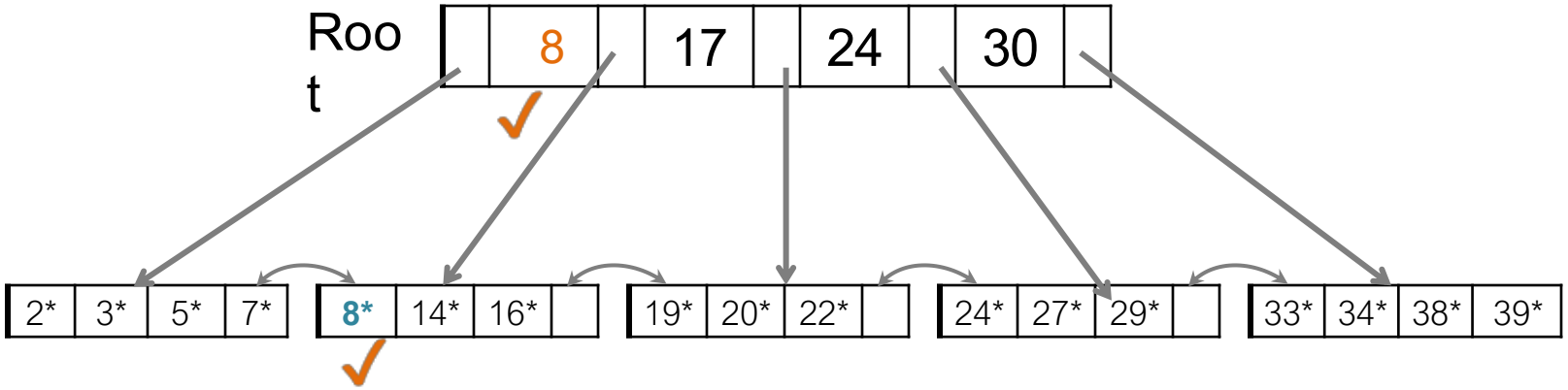


- Re-distributing entries with a sibling
  - + Improves page occupancy, possibly reduces height
  - + May reduce I/O if split would propagate up the tree
  - - Increases I/O (since we check both siblings)
  - Usually not used for non-leaf node splits (cons outweigh pros)
  - Use only for leaf level entries (have to set pointers)
    - No redistribution if next/prev node is full or has diff parent

Reminder: B+-Tree Insertions

- B+-Tree
  - With order d, each node has up to 2*d entries and 2*d+1 pointers (children)
  - Pointer to left of an entry < and pointer to right >=
  - Minimum 50% occupancy (except for root)
  - Height of tree: length of the path from root to a leaf
  - Remains balanced after insert/delete operation.
- Insertion: Find the right leaf
  1. If full, see if (adjacent) sibling has room, redistribute
  2. Otherwise, split the overfull node with (2d + 1) entries into two nodes with d and (d+1) entries, resp.
  3. May have to split parents recursively

B+ Tree Operations

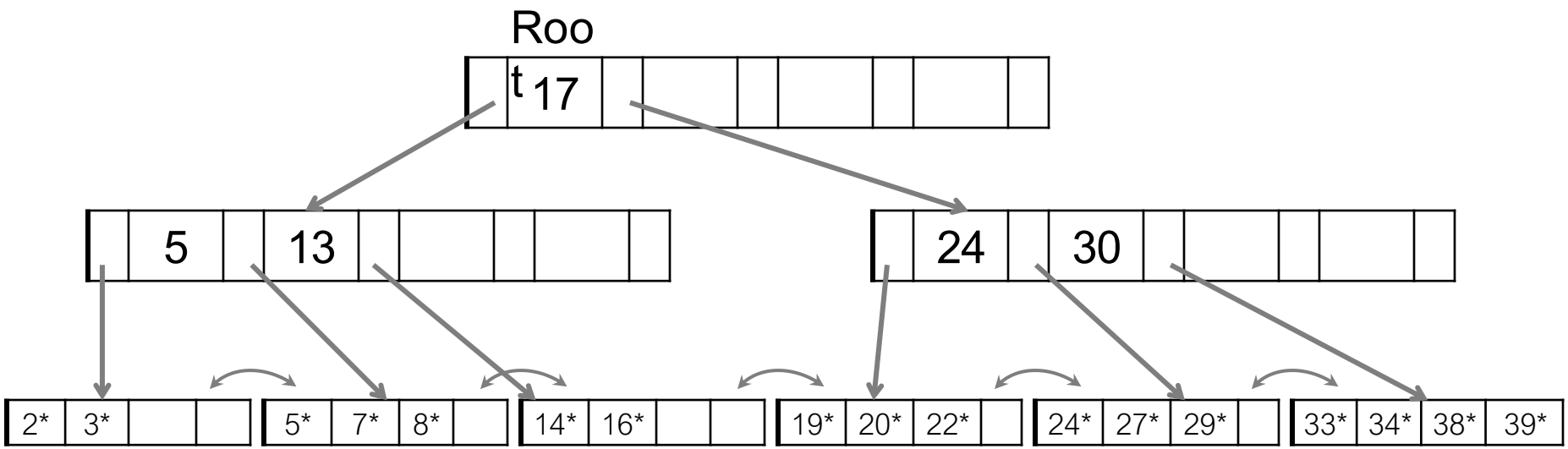
- Search
  - Equality
  - Range
- Insert data entry
- Delete data entry
- Bulk load

B+-Tree: Deleting a Data Entry

1. Find the data entry (will always be at a leaf)
2. Delete it
3. Restore the B+ tree invariant
   - If L is at least half-full, *done!*
   - If L has only d-1 entries,
     - Try to re-distribute, borrowing from a sibling (adjacent node with same parent as L)
     - If re-distribution fails, merge *L* and sibling
4. On merge, delete relevant entry in parent

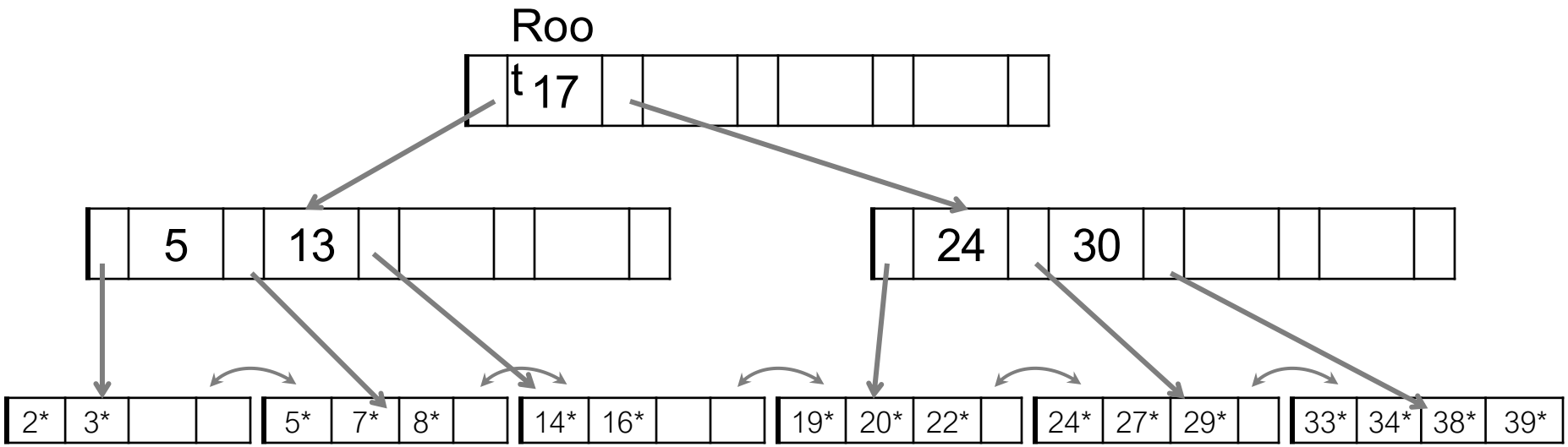
Merge could propagate to root, decreasing height.

Example Tree



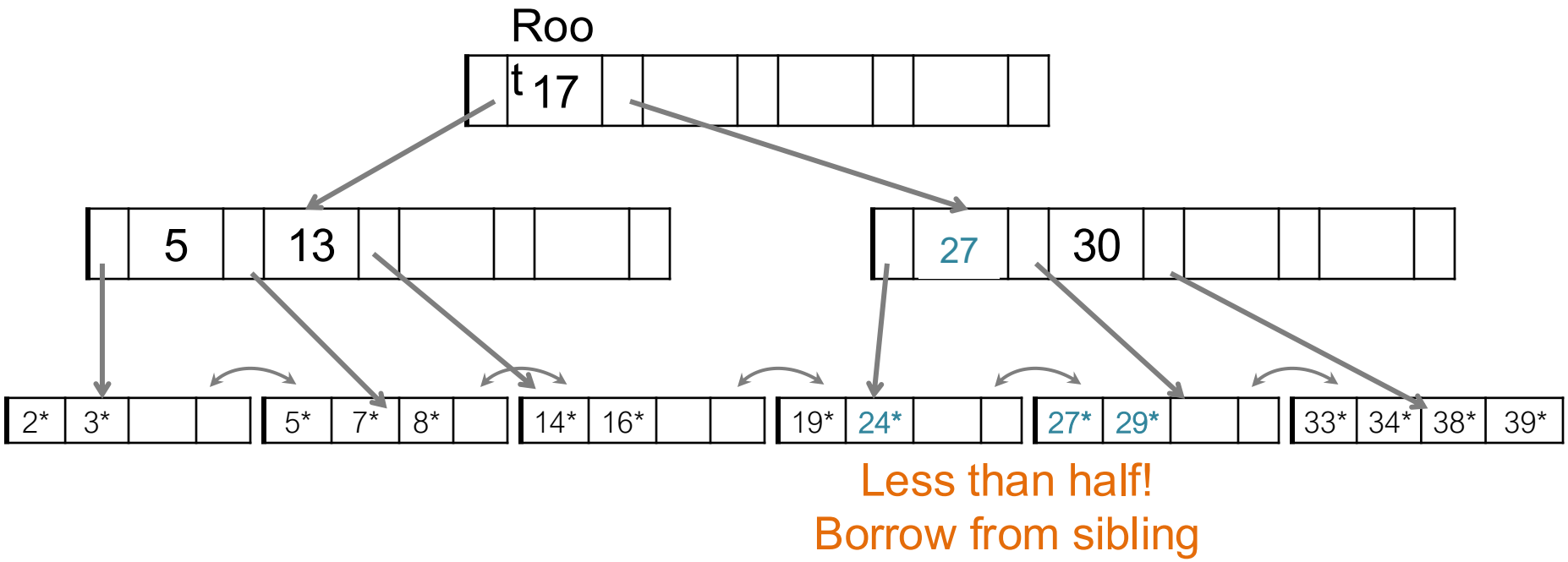
- **Task**: Delete 22, 20, 24
- Deleting 22 is easy. Invariant maintained.
- Deleting 20 is harder. Node would become less than half full.

Deleting 22* and 20*



Less than half!
Borrow from sibling

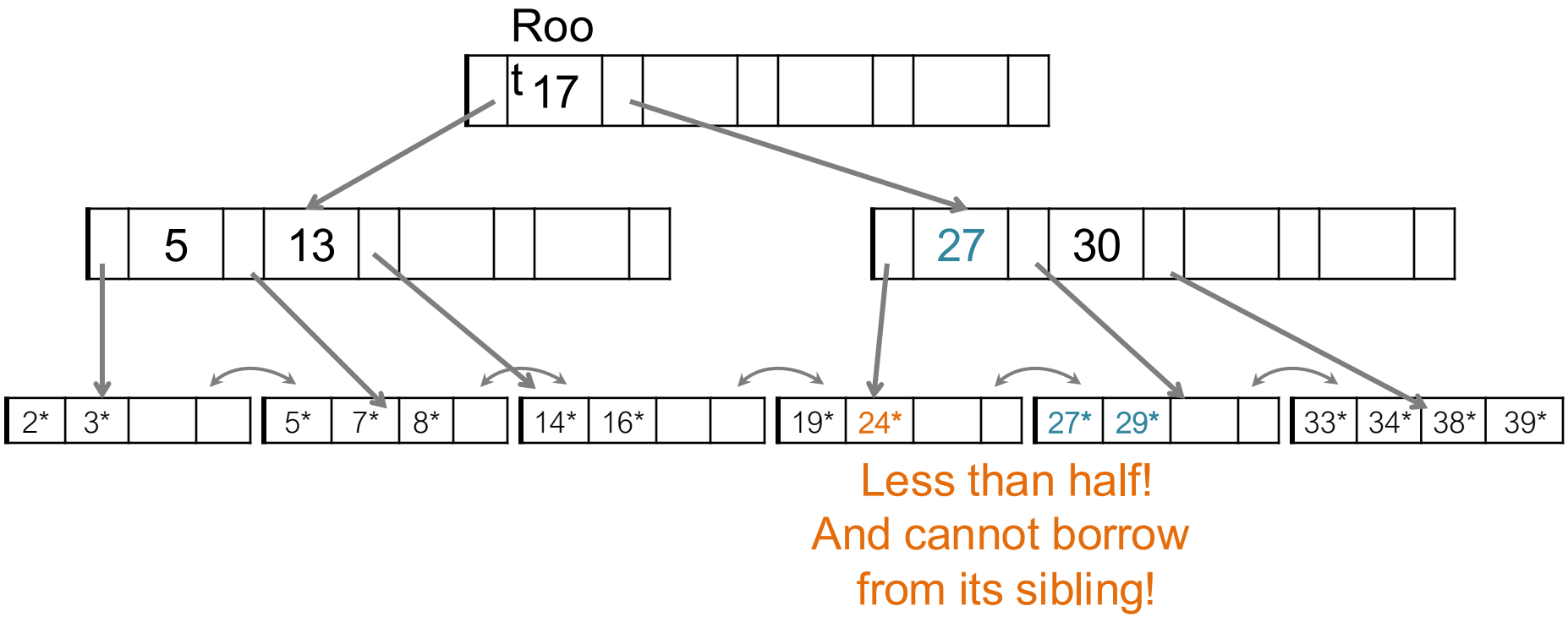
Deleting 22* and 20*



Deleting 20* is done with re-distribution.

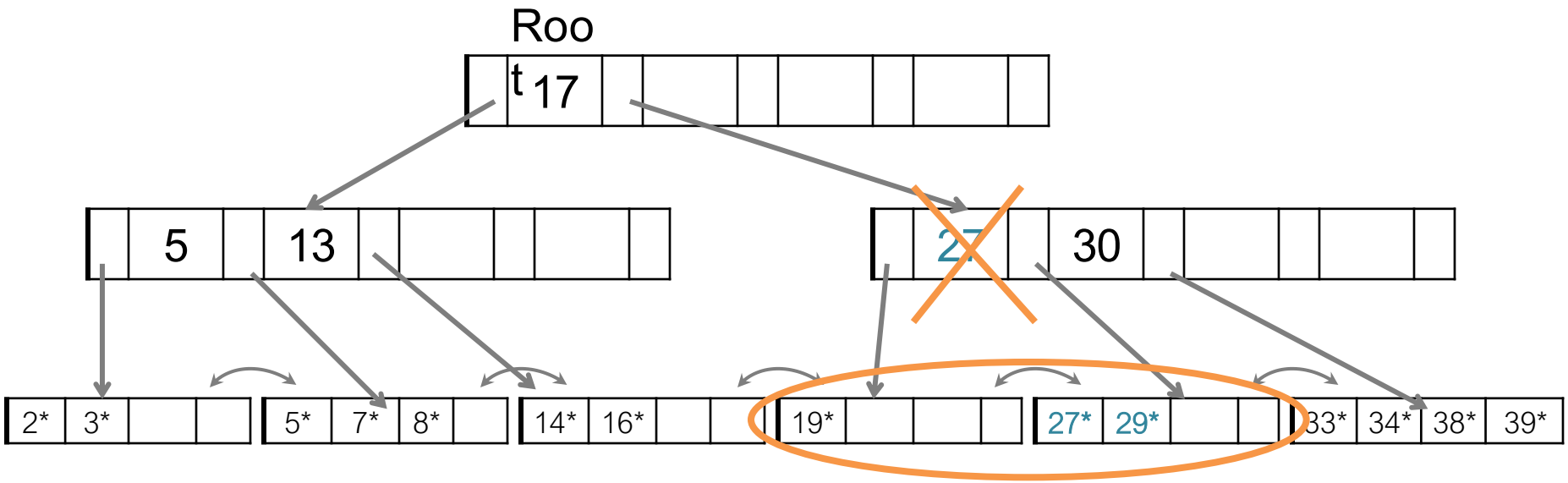
Notice how middle key is copied up.

... And then Deleting 24*



- Must merge
- In the non-leaf node,
  toss the index entry with key value = 27

... And then Deleting 24*



- Must merge
- In the non-leaf node,
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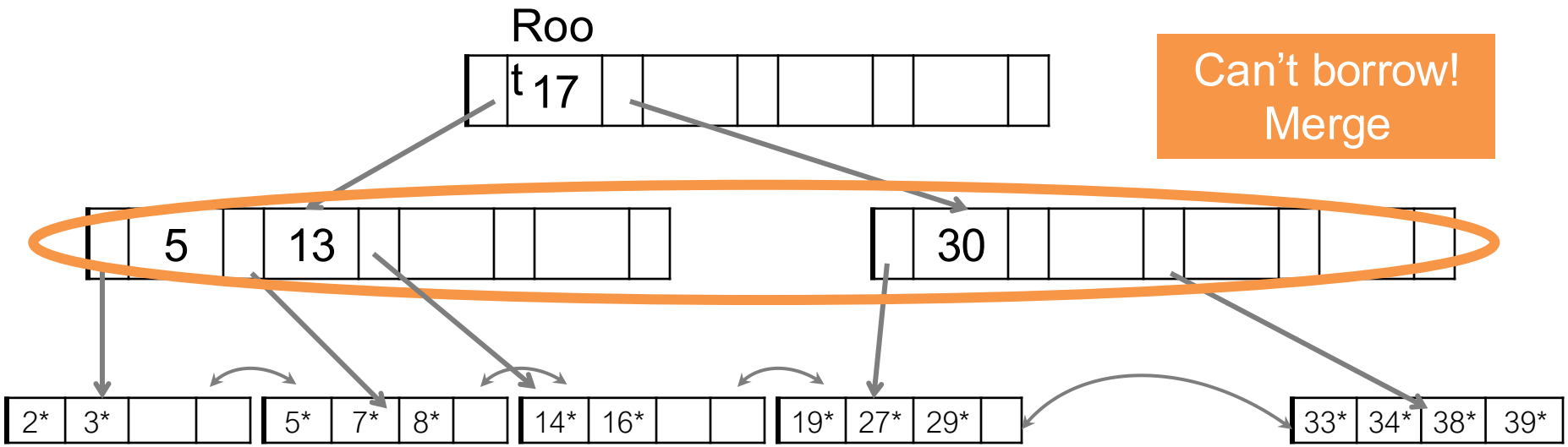
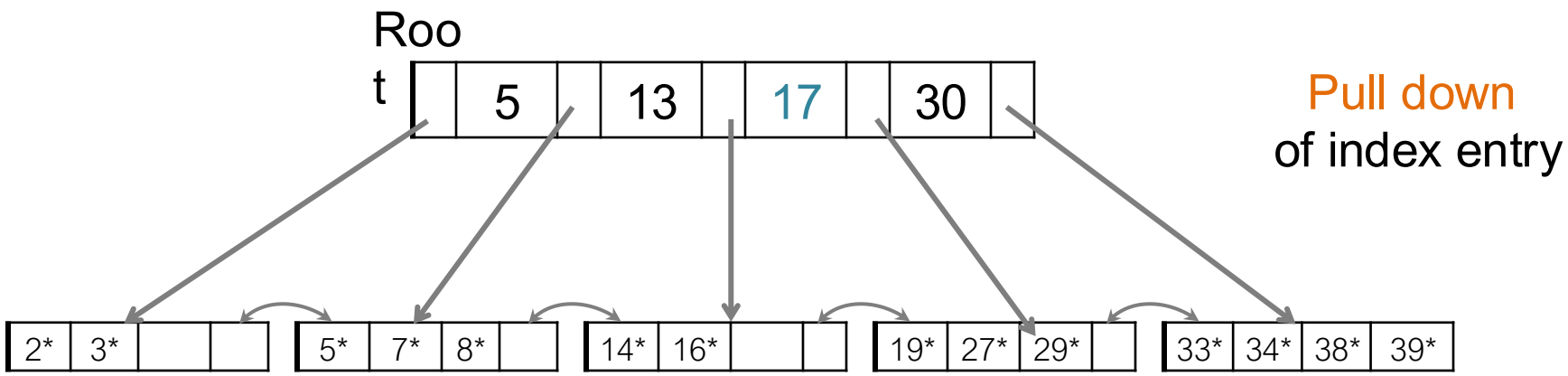
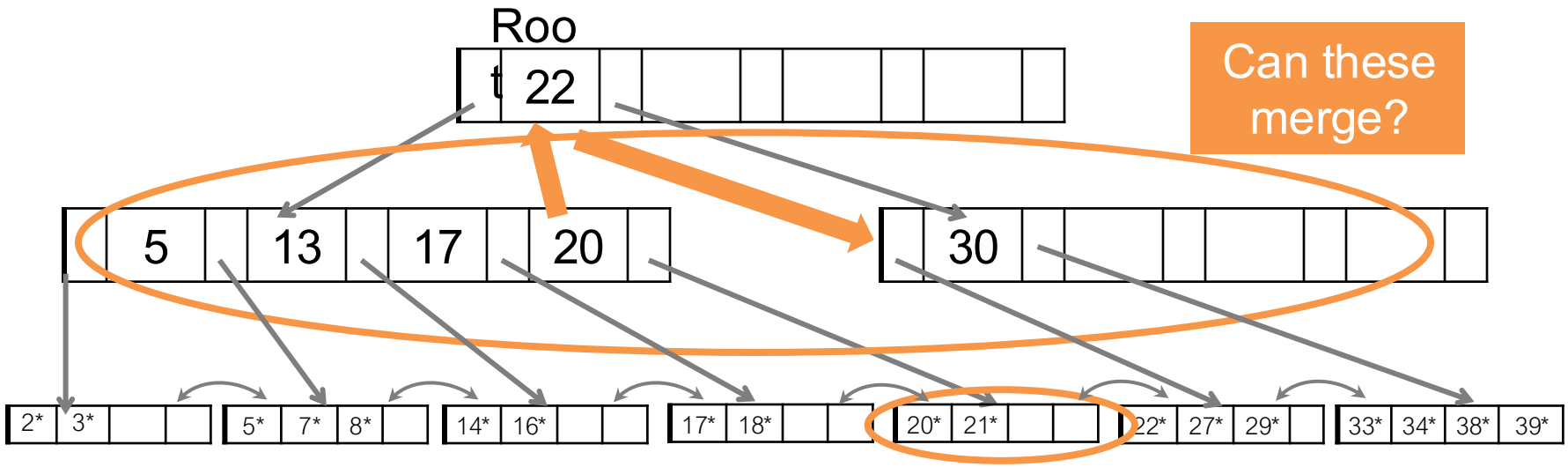
... And then Deleting 24*





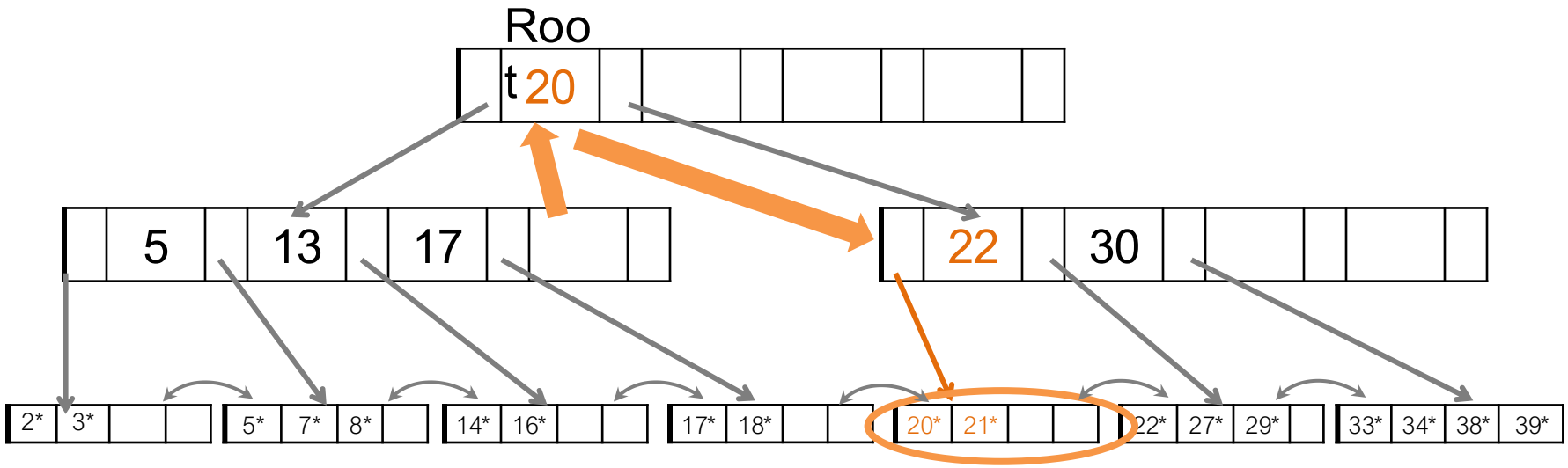
Another Deletion Example: Non-leaf Re-distribution

Can re-distribute entry from left child of root to right child.



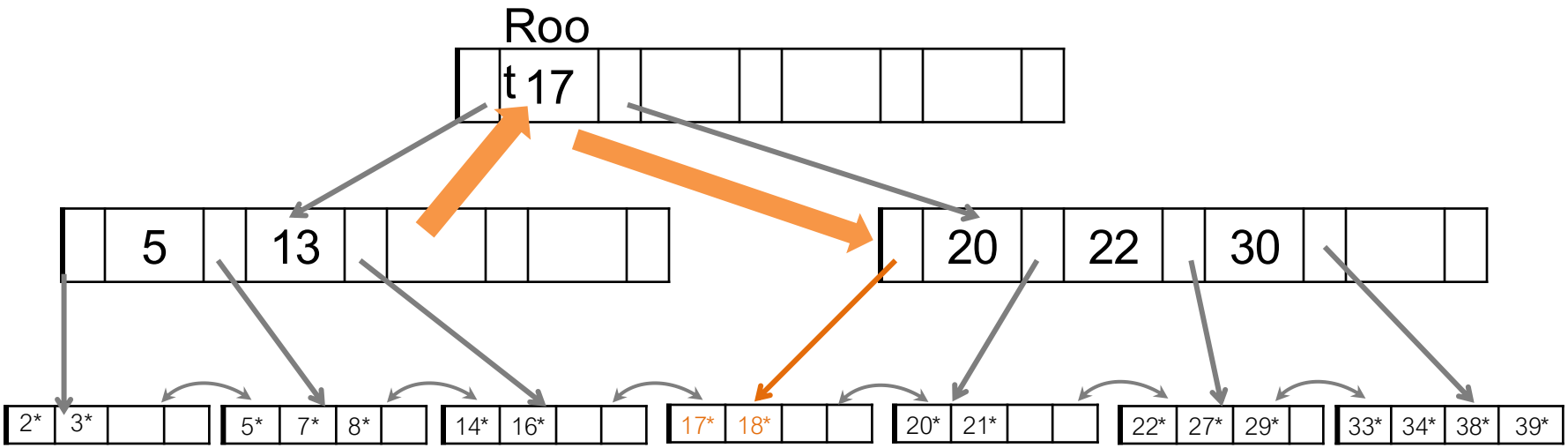
After Re-distribution

- Rotate through the parent node
- re-distribute index entry with key 20



After Re-distribution

It suffices to re-distribute index entry with key 20;
For illustration, here we also re-distribute 17



B+ Tree Deletion

- Try redistribution with all siblings first, then merge. Why?
  - Good chance that redistribution is possible (large fanout!)
  - Only need to propagate changes to parent node
  - Files typically grow, not shrink!

B+ Tree Operations

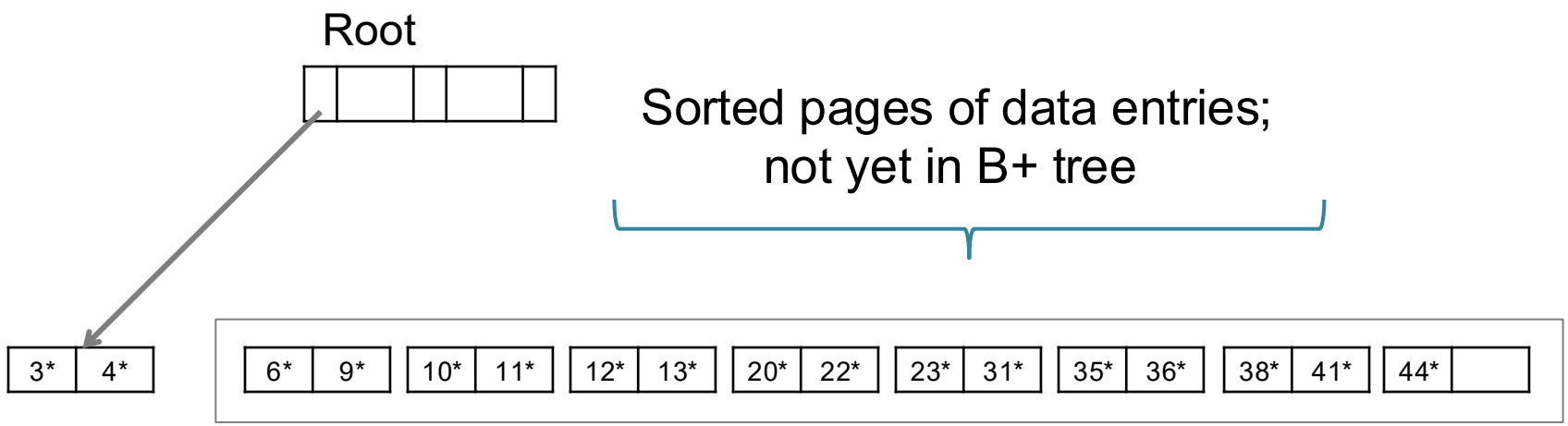
- Search
  - Equality
  - Range
- Insert data entry
- Delete data entry
- Bulk load

Adding Entries to a B+ tree

- Option 1: multiple inserts
  - Slow. Repeated re-organization
  - Does not give sequential storage of leaves
- Option 2: Bulk Loading
  - Fewer I/Os during build
  - Leaves will be stored sequentially (and linked, of course)
  - Can control “fill factor” on pages

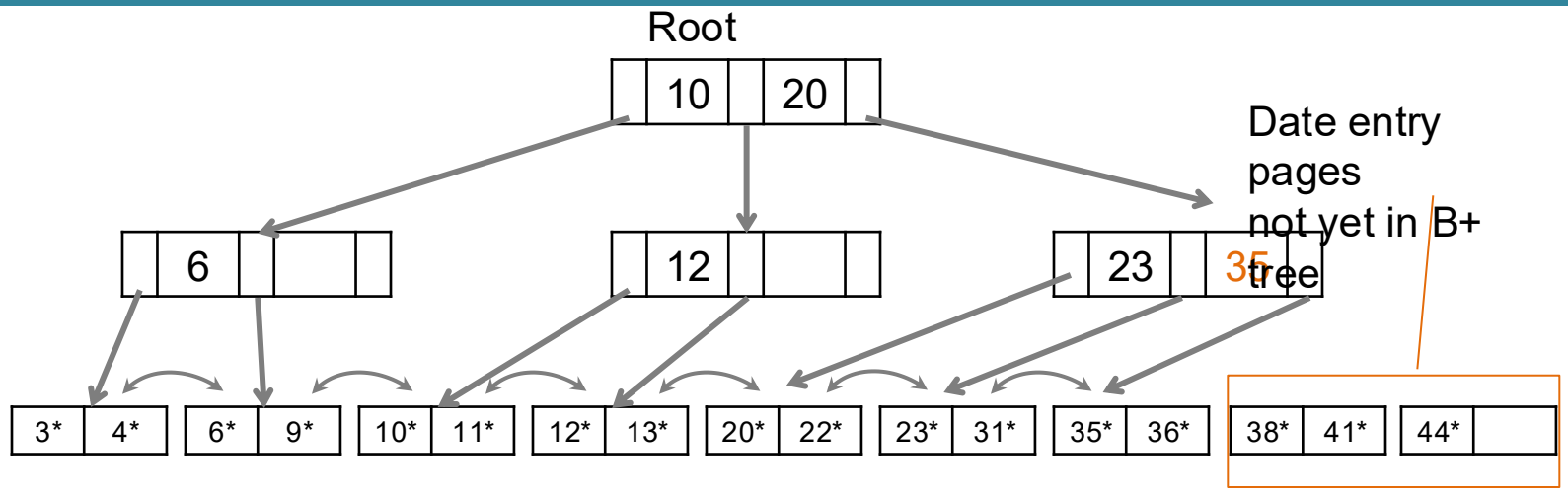
Bulk Loading of a B+ Tree

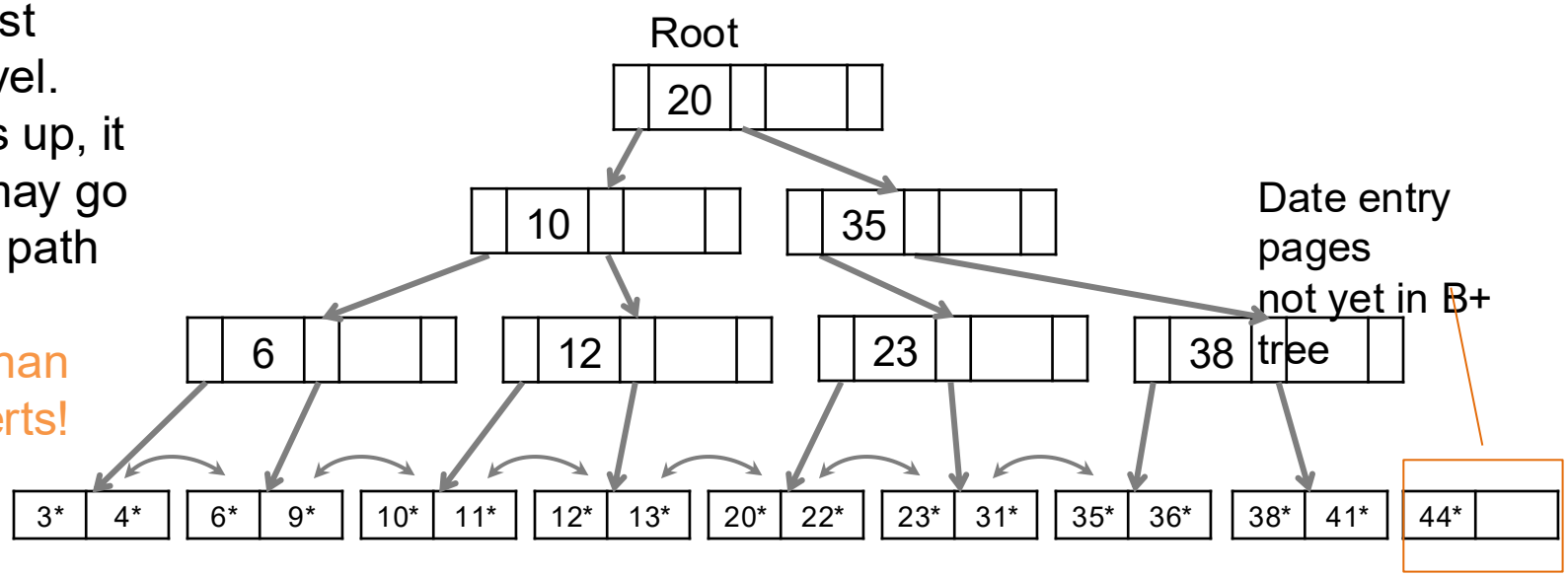
Initialization: Sort all data entries, insert pointer to first (leaf) page in a new (root) page.



Bulk Loading (Contd.)

New index entries for leaf pages always entered into right-most index page just above leaf level. When this fills up, it splits. (Split may go up right-most path to the root.) Much faster than repeated inserts!



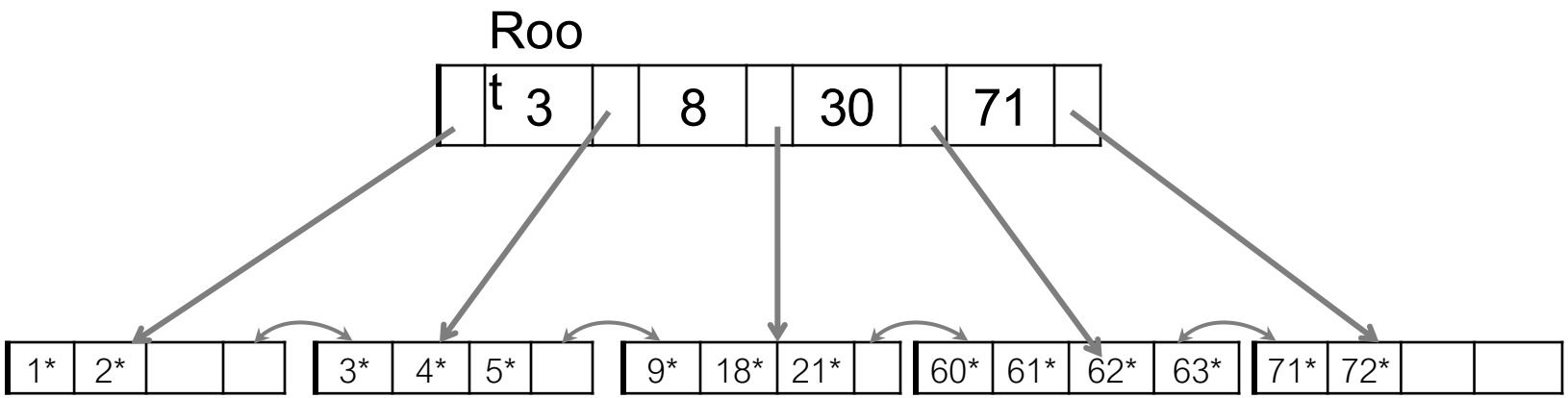


Summary

- Tree-structured indexes are ideal for range searches, also good for equality searches.
- B+ tree is a dynamic height-balanced index structure.
  - Insertions/deletions/search costs $O(\log_F N)$
  - Split: for leaf (copy up), for internal node (push up)
  - Merge: for leaf (toss), for internal node (pull down)
  - High fanout (F) means depth rarely more than 3 or 4.
  - Max occupancy = 2d, Min occupancy = order = d (excp. root)
  - Most widely used index in database management systems because of its versatility. One of the most optimized components of a DBMS.

Exercise

- Ins 6, ins 7, del 21, del 4 (in this order)
- Do not consider re-distribution for insertions (but redistribute for deletions)



Suggested Exercises + Readings

Suggested Exercises: 10.1, 10.5, 10.7

Suggested readings for the lecture on hash-based index:

Read the entire chapter 11 (hash index)

Next…